

The Law of Cosines and the Angle Between Two Vectors

Devi Prasad M

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1 The Law of Cosines

Consider a triangle with side lengths a , b , and c , and an angle θ opposite to side c :

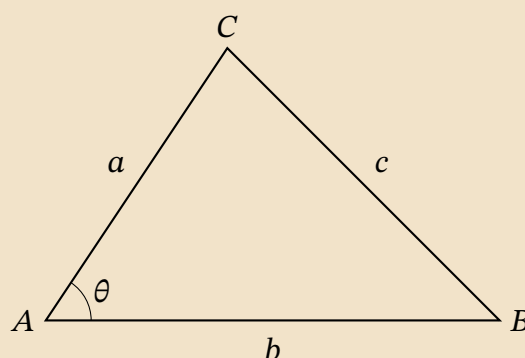


Figure 1: A Generic Triangle.

In such a triangle, the following relationship holds:

$$c^2 = a^2 + b^2 - 2ab \cos \theta \quad (1)$$

1.1 The Geometric Construction

Consider the triangle shown in Figure 1. We drop a height (altitude) h perpendicular to base b from the top vertex, dividing base b into two parts of lengths x and $b - x$.

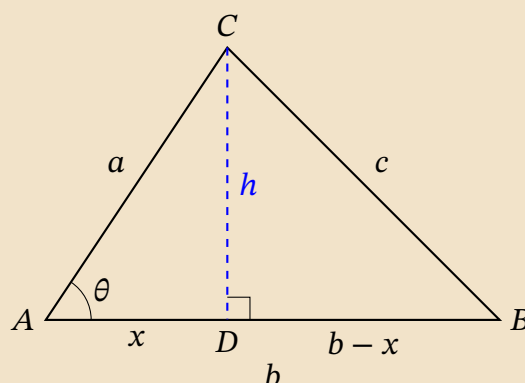


Figure 2: The $\triangle ABC$ with perpendicular h from vertex C to base AB .

2 Proof of the Law of Cosines

Step 1: Express x and h using Trigonometry

Using the right-angled triangle $\triangle ADC$:

$$\cos \theta = \frac{x}{a} \implies x = a \cos \theta \quad (2)$$

$$\sin \theta = \frac{h}{a} \implies h = a \sin \theta \quad (3)$$

Step 2: Apply Pythagorean Theorem

Applying the Pythagorean theorem to the right-angled triangle $\triangle BDC$:

$$c^2 = h^2 + (b - x)^2 \quad (4)$$

Step 3: Substitute and Simplify

Substitute $x = a \cos \theta$ and $h = a \sin \theta$ into equation (4):

$$\begin{aligned} c^2 &= (a \sin \theta)^2 + (b - a \cos \theta)^2 \\ &= a^2 \sin^2 \theta + (b^2 - 2ab \cos \theta + a^2 \cos^2 \theta) \\ &= a^2 (\sin^2 \theta + \cos^2 \theta) + b^2 - 2ab \cos \theta \end{aligned}$$

Using the identity $\sin^2 \theta + \cos^2 \theta = 1$:

$$c^2 = a^2 + b^2 - 2ab \cos \theta \quad \blacksquare \quad (5)$$

3 From the Law of Cosines to the Inner Product

Setting up the correspondence

Let $\mathbf{x}$ and $\mathbf{y}$ be vectors based at the origin A , with $\mathbf{x}$ pointing to C and $\mathbf{y}$ pointing to B , so that

$$a = |\mathbf{x}|, \quad b = |\mathbf{y}|,$$

and θ is the angle between them, exactly as in Figure 1.

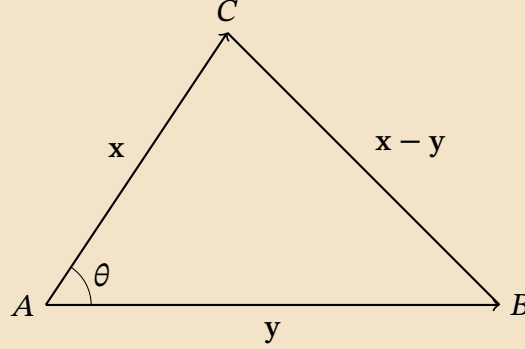


Figure 3: The vectors $\mathbf{x}$ and $\mathbf{y}$, with angle θ between them.

The third side of the triangle is the segment BC , which as a vector is $\mathbf{x} - \mathbf{y}$. Its length is precisely the side c :

$$c = |\mathbf{x} - \mathbf{y}|. \quad (6)$$

Step 4: Expand c^2 algebraically

We use the *algebraic* definition of the inner product, $\mathbf{u} \cdot \mathbf{v} = \sum_i u_i v_i$, together with the identity $|\mathbf{u}|^2 = \mathbf{u} \cdot \mathbf{u}$. Expanding by bilinearity:

$$\begin{aligned} c^2 &= |\mathbf{x} - \mathbf{y}|^2 = (\mathbf{x} - \mathbf{y}) \cdot (\mathbf{x} - \mathbf{y}) \\ &= \mathbf{x} \cdot \mathbf{x} - 2\mathbf{x} \cdot \mathbf{y} + \mathbf{y} \cdot \mathbf{y} \\ &= a^2 - 2\mathbf{x} \cdot \mathbf{y} + b^2. \end{aligned} \quad (7)$$

This is a purely algebraic expression for c^2 — no angle appears in it.

Step 5: Equate to the Law of Cosines

Equation (5) gave us a purely *geometric* expression for the same quantity c^2 :

$$c^2 = a^2 + b^2 - 2ab \cos \theta. \quad (8)$$

Since (7) and (8) both equal c^2 , we may set them equal to each other:

$$a^2 - 2\mathbf{x} \cdot \mathbf{y} + b^2 = a^2 + b^2 - 2ab \cos \theta.$$

Step 6: Cancel and solve

The terms a^2 and b^2 appear on both sides and cancel:

$$-2 \mathbf{x} \cdot \mathbf{y} = -2ab \cos \theta.$$

Dividing both sides by -2 :

$$\mathbf{x} \cdot \mathbf{y} = ab \cos \theta = |\mathbf{x}| |\mathbf{y}| \cos \theta. \quad \blacksquare \tag{9}$$

Remark

The Law of Cosines was proved purely from Pythagoras, with no reference to inner products.

Equation (7) was proved purely from the algebraic (coordinate) definition of the inner product, with no reference to angles.

The bridge between algebra and geometry is the single quantity $c = |\mathbf{x} - \mathbf{y}|$: one formula for c^2 comes from coordinates, the other from Pythagoras, and equating them is precisely what encodes $\cos \theta$ inside the inner product.